Project Design Phase

Solution Architecture

Date	12-10-2026
Team ID	SWUID2026402745
Project Name	HotelPro - Hotel Room Booking and Management System
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for **HotelPro** ensures a robust, secure and scalable platform that connects travelers and guests with hotel properties. It follows a layered three-tier design: a React.js single-page application (SPA) with Bootstrap responsive UI, a stateless Node.js and Express.js REST API secured with JSON Web Tokens (JWT) and role-based access control (RBAC), and a MongoDB document database utilizing Mongoose ODM, supported by database indexing and validation. The architecture focuses on responsive design, real-time availability checking, double-booking prevention, secure payment gateway integration and comprehensive hotel inventory and guest management.

- Seamless hotel and room discovery, search filtering by city, dates, guest capacity, price, room type and amenities
- Real-time room availability checking with confirmation-time re-verification and compound indexing to eliminate double bookings
- Scalable user authentication and role-based authorization (Customer, Hotel Staff/Receptionist, Administrator) using JWT
- Secure online payment processing (credit/debit cards, UPI, net banking) via an integrated payment gateway alongside offline counter payment recording and automated invoicing
- Digital guest record management, front-desk check-in/check-out processing, room status updates and actionable admin dashboard analytics with CSV/PDF reporting

Example - Solution Architecture Diagram:

